

RAJ ARYAN

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EXPERIENCE

Software Development Engineer (SDE) Intern

Umeed Social and Economic Development Date: May 2026 – July 2026

Location: Varanasi, India

Python LangChain ML Models ChromaDB REST APIs

- Built and maintained **backend services** for the organization's internal data platform, designing modular **REST APIs** for reliable data ingestion and retrieval.
- Fine-tuned and deployed **ML models** for domain-specific NLP tasks, applying transfer-learning techniques to improve inference accuracy on organizational data.
- Researched and processed unstructured organizational data using **LangChain** and vector-based retrieval techniques, collaborating with the **Project Coordinator** to validate R&D findings.

PROJECTS

IRCTC Backend (GitHub)

Microservices-based Railway Booking System

Node.js Express.js Kafka PostgreSQL Redis

- Architected a production-grade microservices backend with **9** independently deployable services using Node.js, Express.js, PostgreSQL, and Prisma ORM.
- Implemented secure **JWT** authentication with OTP/Google OAuth and engineered Redis-based distributed seat-locking to prevent race conditions during concurrent bookings.
- Built an event-driven booking-payment Saga on **Apache Kafka** across **13+** topics, with an API Gateway, rate limiting, and Elasticsearch-powered search, containerized via Docker Compose.

Local Generative AI RAG Application (GitHub)

Retrieval-Augmented Generation System

Python LangChain ChromaDB Streamlit LLMs

- Architected a **100%** offline RAG system using Python and LangChain, delivering privacy-preserving document analysis via **ChromaDB** vector search.
- Engineered an NLP pipeline to chunk and embed PDFs with HuggingFace models, and deployed local **LLMs** via Ollama (DeepSeek-R1) with real-time token streaming.
- Built a fault-tolerant **Streamlit** UI with dynamic hyperparameter tuning, ensuring zero-crash application stability.

Opportune (GitHub)

Full-Stack Job Portal

MERN Redux Toolkit Tailwind CSS Cloudinary

- Architected a full-stack job portal on the **MERN** stack with role-based access control for job seekers and recruiters.
- Engineered secure REST APIs with **JWT** authentication and bcrypt hashing, integrating Cloudinary for scalable resume/logo uploads.
- Built a dynamic frontend with React, Redux Toolkit, and Tailwind CSS featuring an applicant tracking dashboard.

EDUCATION

IIIT Allahabad

Aug '23 – Jun '27

B.Tech in Information Technology

CGPA: 7.37 (6th Semester)

Greenford International Academy

Class 12th

Jul '21 – Jun '22

Score: 86%

Genesis International School

Class 10th

Aug '19 – Jun '20

Score: 90%

TECHNICAL SKILLS

Relevant Coursework

Data Structures and Algorithms OOP

Computer Networks Operating Systems

Artificial Intelligence Machine Learning

Generative AI

Computer Science Fundamentals

Languages

JavaScript (ES6+) HTML5 CSS3

C/C++ Java React Node.js

Express.js MongoDB

Developer Tools

VS Code Postman Git IntelliJ

Technologies and Tools

REST APIs Microservices Architecture

JWT Authentication OAuth Apache Kafka

Redis PostgreSQL Prisma ORM

Elasticsearch Docker Docker Compose

Redux Mongoose Agile Methodologies

Generative AI

LINKS

- Portfolio Link: Portfolio-website

ACHIEVEMENTS

- Team secured **4th position** in OOC, IIITA Design Hackathon, out of **103** teams.
- Secured **Rank 440** in CodeChef's **Monday Munch - DSA Challenge 016**.

RESPONSIBILITIES

- **Member, Public and Relations** – Aproxsha'24
- **Member** – Prayaas (NGO, IIITA)
- **Member** – IIC (IIITA Incubation)